Yosuke HIGUCHI

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Education

Waseda University — Tokyo, Japan
M.S. in Computer Science and Engineering — *Apr. 2019 - Present*
- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research Field: Speech Recognition, Machine Learning

Waseda University — Tokyo, Japan
B.E. in Computer Science and Engineering — *Apr. 2015 - Mar. 2019*
- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research Field: Speech Recognition, Machine Learning

Research Topic

Zero-resource speech recognition — Waseda University
Mentor: Naohiro Tawara, Tetsuji Ogawa, Tetsunori Kobayashi — *Jun. 2018 - Mar. 2019*
- Proposed a framework for extracting speaker-invariant features from zero-resource languages. The framework consists of two phases: unsupervised clustering to obtain phonetic information and domain-adversarial training to suppress a variety in speakers.

Non-parallel many-to-many voice conversion with StarGAN — NTT CS Laboratory
Mentor: Hirokazu Kameoka — *Aug. 2018 - Sept. 2018*
- Worked on improving a StarGAN-based voice conversion model. Investigated different types of neural network architecture and designed a new objective function to further improve the previous model.

Speaker embeddings for speaker diarization — IBM Research AI
Mentor: Masayuki Suzuki, Gakuto Kurata — *Jun. 2019 - Oct. 2019*
- Proposed methods of training speaker embeddings for speaker diarization. The embeddings are trained to incorporate differences in acoustic conditions between recording environments.

Noise-robust end-to-end speech recognition — Waseda Univ./NTT CS Lab.
Mentor: Naohiro Tawara, Tomoharu Iwata, Atsunori Ogawa, Tetsuji Ogawa — *Apr. 2019 - Dec. 2019*
- Proposed a training framework for improving the noise robustness of the end-to-end speech recognition network. The attention weights estimated from a clean speech are used to approximate the weights from the noisy counterpart.

Non-autoregressive end-to-end speech recognition — Johns Hopkins University
Mentor: Shinji Watanabe — *Dec. 2019 - Mar. 2020*
- Proposed a non-autoregressive model for end-to-end speech recognition. The model generates a sequence by refining the CTC outputs based on mask prediction.

Work Experience

InfoDeliver Corporation — Tokyo, Japan
Part-time Software Engineer — *Sept. 2016 - Jun. 2017*
- Involved in developing and maintaining web services and mobile apps.

NTT Communication Science Laboratories, NTT Corporation — Kanagawa, Japan
Research Internship — *Aug. 2018 - Sept. 2018*
- Research Topic: voice conversion, generative adversarial network

IBM Research AI — Tokyo, Japan
Research Internship — *Jun. 2019 - Oct. 2019*
- Research Topic: speaker diarization

Johns Hopkins University Baltimore, MD

Research Internship/Visiting Scholar *Dec. 2019 - Mar. 2020*

- Research Topic: end-to-end speech recognition

Publication

Refereed Conference

- **Yosuke Higuchi**, Naohiro Tawara, Tetsunori Kobayashi, Tetsuji Ogawa, "Speaker Adversarial Training of DPGMM-based Feature Extractorfor Zero-resource Languages," In *Proceedings of 20th Annual Conference of International Speech Communication Association (INTERSPEECH)*, September 2019.
- **Yosuke Higuchi**, Masayuki Suzuki, Gakuto Kurata, "Speaker Embeddings Incorporating Acoustic Conditions for Diarization," In *Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing* (ICASSP), May 2020. (to appear)
- **Yosuke Higuchi**, Shinji Watanabe, Nanxin Chen, Tetsuji Ogawa, Tetsunori Kobayashi, "Mask CTC: Non-Autoregressive End-to-End ASR with CTC and Mask Predict," In *Proceedings of 21th Annual Conference of International Speech Communication Association (INTERSPEECH)*, October 2020. (to appear)
- **Yosuke Higuchi**, Naohiro Tawara, Tomoharu Iwata, Atsunori Ogawa, Tetsunori Kobayashi, Tetsuji Ogawa, "Noise-robust Attention Learning for End-to-End Speech Recognition," In *28th European Signal Processing Conference (EUSIPCO)*, January 2021. (to appear)

Domestic Conference/Workshop

- **Yosuke Higuchi**, Naohiro Tawara, Tetsuji Ogawa, Tetsunori Kobayashi, "Speaker Invariant Feature Extraction for Zero-Resource Languages," In *2019 Spring Meeting of Acoustical Society of Japan (ASJ)*, March 2019.
- **Yosuke Higuchi**, Naohiro Tawara, Tetsunori Kobayashi, Tetsuji Ogawa, "Speaker Adversarial Training of DPGMM-based Feature Extractorfor Zero-resource Languages," In *IPSJ SIG Technical Reports on Spoken Language Processing (IPSJ-SLP)*, September 2019. [best student paper]
- **Yosuke Higuchi**, Naohiro Tawara, Tomoharu Iwata, Atsunori Ogawa, Tetsunori Kobayashi, Tetsuji Ogawa, "Noise-robust Attention Learning for End-to-End Speech Recognition," In *2020 Spring Meeting of Acoustical Society of Japan (ASJ)*, March 2020.
- **Yosuke Higuchi**, Masayuki Suzuki, Gakuto Kurata, "Speaker Embeddings Incorporating Acoustic Conditions for Diarization," In *2020 Autumn Meeting of Acoustical Society of Japan (ASJ)*, September 2020. (to appear)
- **Yosuke Higuchi**, Shinji Watanabe, Nanxin Chen, Tetsuji Ogawa, Tetsunori Kobayashi, "Mask CTC: Non-Autoregressive End-to-End ASR with CTC and Mask Predict," In *2020 Autumn Meeting of Acoustical Society of Japan (ASJ)*, September 2020. (to appear)

Skills

Natural language	Japanese (native), English (TOEFL iBT: 102, TOEIC: 950)
Programming language	Python, C, C++, Java, JavaScript, LaTeX
Software & Tools	PyTorch, Chainer, ESPnet, Kaldi